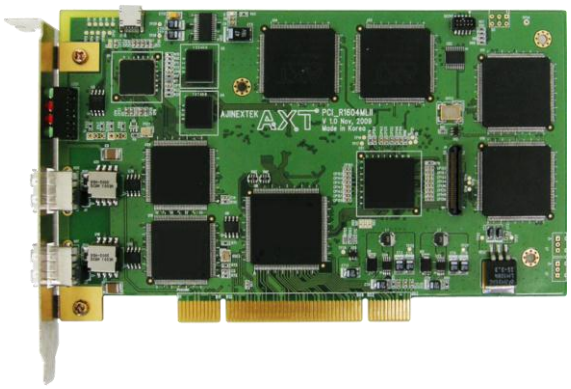
PCI-R1604-MLII MLII Master Board



PCI-R1604-MLII is a master board for MLII that allows the control of up to 30 slave nodes through the MLII(MeChaTrolINk-II) communication protocol. Of the 30 Slave nodes, a standard of 16 motion function modules (Servo and Pulse Output Modules for MLII) may be installed and motion function modules may be installed to all slave nodes, a total of 30, with the installation of option board.
A set of example programs with 200 motion functions that helps with making various control programs in a set of different languages is provided.

- Mechatrolink-II (YASKAWA standard of distributed network)
- Standard 16Axes(Max. 30 Axes Expandable)
- Max. 30 MLII Slave Module Control
- For each 1.0 mSec all information of 30 modules are recorded
- Motion Synchronization Drive (Support for max. 30 Synchronization Drive, 2~4 Axes Linear Interpolation, 2 Axes Circular Interpolation, Continuous Interpolation, 3 Axes Helical Interpolation)

Product Feature

Number of Dynamic Axes

Standard 16 Axes control (20, 24, 28, 30 Axes control possible with expansion)

Communication Protocol

Mechatrolink-II (YASKAWA standard of distributed network)

Network Connection

Exclusive STP Cable for Network Connection (C.I. : 130Ω), Min.: 0.5m(Between Nodes), Max.: 50m(For each Network)

Synchronization Drive

Axes: Max. 30 Axes Synchronization Drive, 2-4 Axes Linear / 2 Axes Circular Interpolation, 3 Axes Helical Interpolation

Drive Method

Fixed Distance Drive, Jog Drive, Signal Detection Drive, Interpolation Drive, ECAM Drive

Override Function

Function Support for Modification to Speed and Distance in Operation (Drive)

Various Adjustable Speed Mode

Support for Symmetrical Trapezoid, Symmetrical S-Curve, Asymmetrical Trapezoid, Asymmetrical S-Curve Drives

Motion Instruction Speed

Max. Motion Position Instruction Modification Speed: ~10.0000Mpps

Advanced Motion Drive

Servo Driver Control Speed Drive/Torque Drive Function, Real-time Position Proofing Function, ECAM Function

Motion Position Range

32 Bit Range Count (Same for Command / Actual)

Supported Slaves

SGDV-XXXX11/15 Servo Drive of YASKAWA, JEPMC-2910 Pulse Output Module, JEPMC-IO2310 Digital Input and Output Module, JEPMC-AN29x0 Analog Input and Output Module

Bus Connection Method

PCI 2.1 Memory Access, 32bit

Support Development Environment

OS: Microsoft Windows 2000/XP/7(32bit, 64bit)

Development Tool: Visual-C++, Delphi, C-Builder, Visual-Basic, C#

Development Environment: RTX kernel library Provided

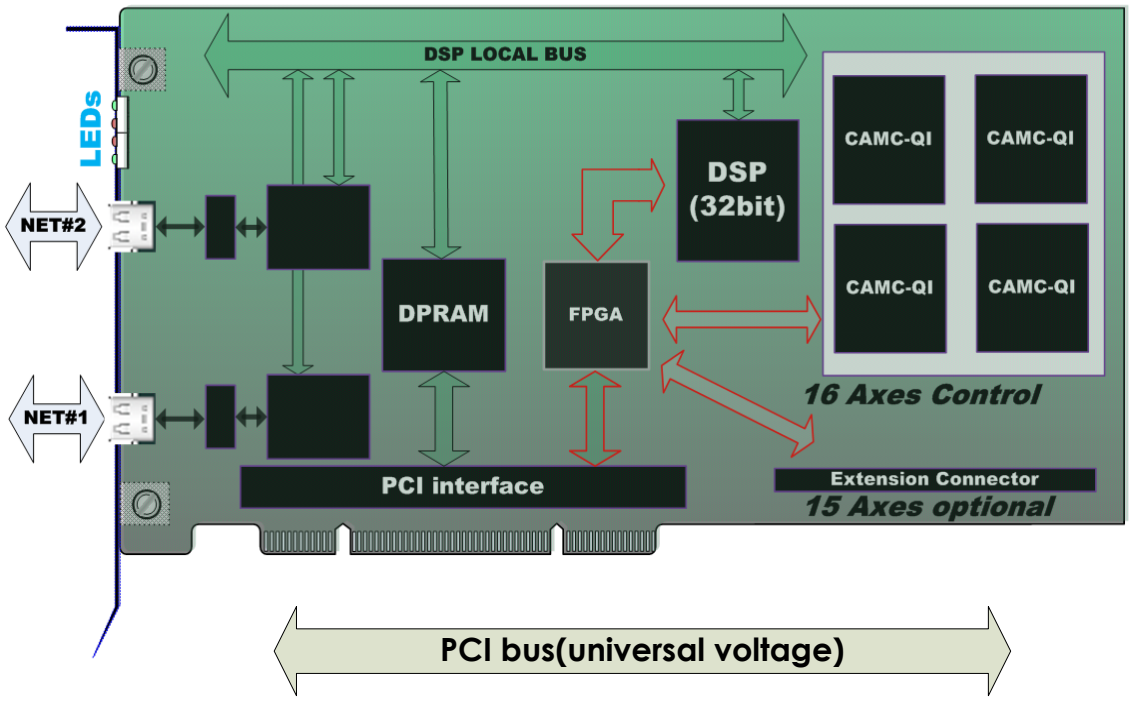
Product Specification

Items	Specification
Communication type	Mechatrolink-II (Yaskawa standard of distributed network)
MLII Port	2 port(15 Slaves installable in each port)
Control Axis	Standard 16 Axes, Max. 30 Axes installable with the option board
Motion Control chip set	CAMC-QI 4EA
Velocity of motion command	~10.0Mpps Command Position change rate (Clock 39.3216MHz standard)
LED display	Each Port : Comm(Green), Error(Red)
Number of configuration slave	Max 30 MLII Slave Module
Data Bus Width	32-bit Range Counter(command/actual equal)
Bus Interface	PCI 2.2, Memory Access, 32bit
Current consumption	Max 1.0A (5Vdc)
Operating temperature / Humidity	Temperature: 0 ~ 50°C / Humidity: below 80%
Storage temperature	-20 ~ 70°C
Weight	30g (172g when expanded to 30 axes)
Exterior dimension of product	174.63 (W) mm x 106.00 (L) mm

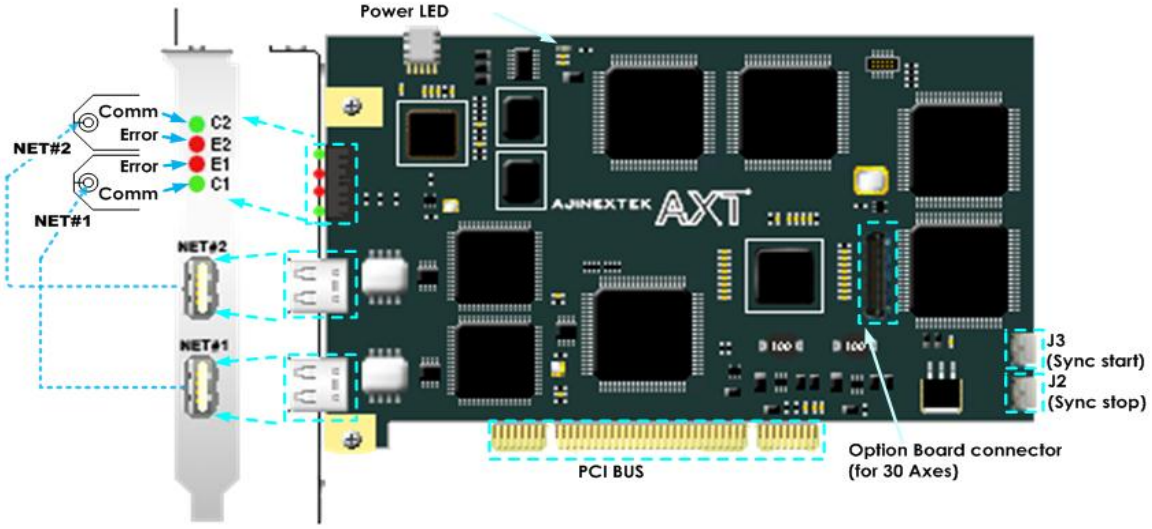
Support Software

Items	Specification
Operating System	Windows 2000, XP, 7(32/64 bit)
User Agent Software	EzSoftware RM
Support Language	Visual C++, Visual Basic, Delphi, C++ Builder, C#, RTX kernel

Hardware Block Diagram



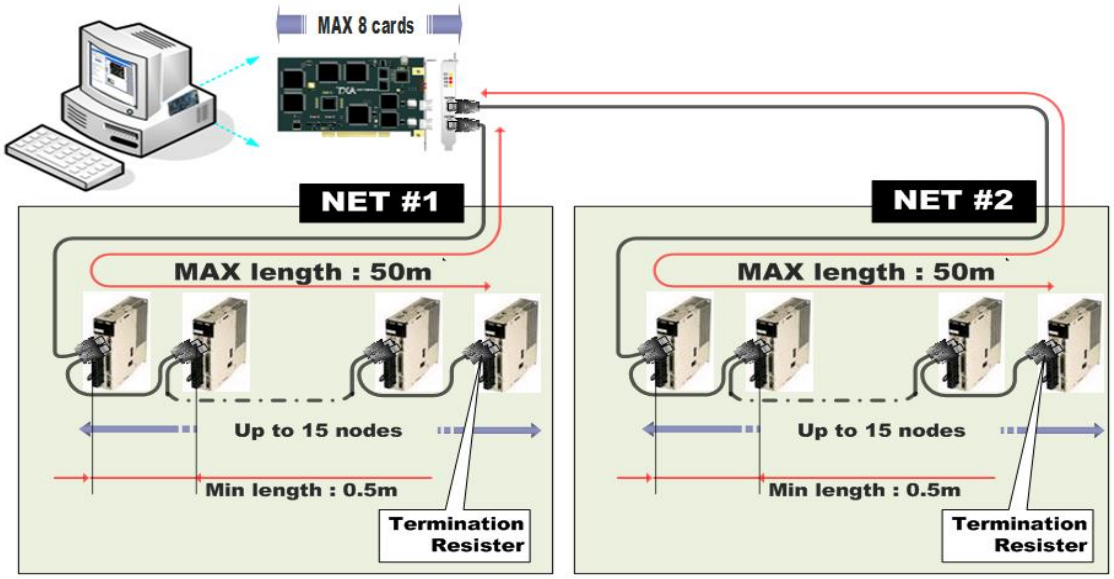
Connector and LED information



Related Accessory

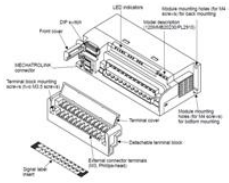
Product	Model Name
Cable(0.5m ~ 30m)	JEPMC-W6002-xx(Mechatrolink-II Cable, USB to USB)
Terminator (terminating resistance)	JEPMC-W6022(Must be connected at the last Slave Node)

MLII Network System



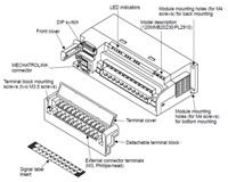
PCI-R1604-MLII can install a maximum of 30 slave modules with 15 modules for each NET and also install a standard of 16 motion function modules (SGDV-xxxx11A).

Slave Module



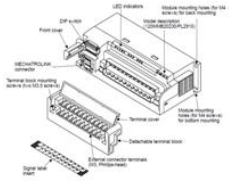
JEPMC-AN2900

A/D Module Analog input -10 ~ 10V, 2 channels



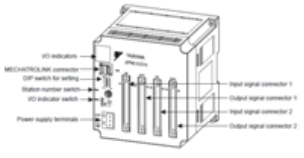
JEPMC-AN2910

D/A Module Analog output -10 ~ 10V, 2 channels



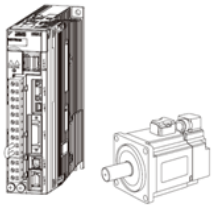
JEPMC-PL2910

Pulse Output Module Pulse output, 2 channels



JEPMC-IO2330

64-point Module 24VDC, 64-point input, 64-point output(sourcing)



SGDV-xxxx11A(Σ – V MLII Series)

Servo driver for MLII, Motor(SGMJV, SGMAV, SGMPS, SGMGV, SGMSV, SGMCS)